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Roll No.

220931

3rd Sem / Electrical
Subject : Electrical machines -I

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Electrical machine which converts mechanical energy into electrical energy is known as (CO1)

- a) Electrical generator
- b) Electrical motor
- c) Transformer
- d) All of the above

Q.2 Which of the following motors has high starting torque. (CO2)

- a) DC shunt motor b) DC series motor
- c) Both d) None of the above

- Q.3 Transformer are rated in (CO3)
- a) KW b) KV
 - c) KWH d) KVA
- Q.4 Which of the following is not a part of transformer (CO4)
- a) Conservator b) Breather
 - c) Buchholz relay d) Exciter
- Q.5 Which type of connection in a three phase transformer is used for the substation end of the transmission line? (CO4)
- a) Star/star b) Delta/Delta
 - c) Star/Delta d) Delta/Star
- Q.6 Auto transformer has
- a) One winding
 - b) Multiple winding
 - c) Two winding
 - d) Does not have winding

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 The efficiency of a transformer will be maximum when variable losses = _____ (CO3)

- Q.8 DC series motor is a _____ speed motor. (CO2)
- Q.9 The open circuit test of transformer is used to measure the _____ losses. (CO3)
- Q.10 In step down transformer, primary turns are _____ than secondary turns. (CO3)
- Q.11 A transformer has no _____ losses. (CO3)
- Q.12 The auto-transformer requires _____ Copper than a two winding transformer of same rating. (CO5)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Discuss the losses in a DC machine? (CO1)
- Q.14 Derive the EMF equation of DC generator. (CO1)
- Q.15 Differentiate between a generator and motor. (CO1)
- Q.16 Write the methods of speed control of DC shunt motor. Explain any one? (CO2)
- Q.17 What are the conditions for parallel operation of 3-phase transformer. (CO4)
- Q.18 Draw and explain the short circuit test on the single-phase transformer. (CO3)
- Q.19 Draw a phasor diagram of 1-phase transformer for inductive load. (CO3)

- Q.20 What is an auto-transformer and what are its advantages and disadvantages. (CO5)
- Q.21 Can a transformer work on dc? Explain. (CO3)
- Q.22 Explain the construction and working of Instrument transformer. (CO5)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Explain the construction, principle and working of 1-phase transformer with neat sketch. (CO3)
- Q.24 Explain the armature reaction of DC machine. (CO1)
- Q.25 Draw and explain the connection of various types of three-phase transformer. (CO5)